



Faculty of Computer Science and Engineering

Department of Cyber Security

Wireless and Mobile Security

Course Code: CY315

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Security Threats Overview: Wired, Wireless and Mobile

Lecture Goals

- ❖ **Identify general threat categories**
- ❖ **Explain C-I-A-A-N principles**
- ❖ **Recognize wireless/mobile vulnerabilities**
- ❖ **Understand authentication vs authorization**
- ❖ **Explain risk mitigation approaches**

What to Protect?

What Are We Protecting?

- ❖ Before designing security, ask:
 - What are the assets?
 - Why protect them?
 - What is their value?
 - What are the threats?
 - What are the constraints?
- ❖ Security must be **cost-effective**.

Types of Assets

❖ **Assets may include:**

- Corporate data
- Intellectual property
- Customer data
- Financial systems
- Remote-controlled IoT devices

❖ Different assets require different protection levels.

Business vs Security Conflict

❖ **Security officers want:**

- Tight controls
- Restricted access
- Minimal exposure

• **Business leaders want:**

- Easy access
- High productivity
- Mobility

❖ Security measures must be proportional to asset value.

Least Privilege Principle

Organizations implement **Least Privilege**

❖ **Users receive:**

- Only permissions required
- No unnecessary access

❖ **This reduces:**

- Insider threat exposure
- Accidental misuse

Security Must Align with Business

❖ You cannot:

- Disconnect e-commerce servers from Internet
- Disable mobile access completely

❖ Security must:

- Enable authorized business
- Prevent unauthorized activity

❖ Security should not become an obstacle.

Expanded Security Principles

Beyond the C-I-A Triad

❖ Traditional C-I-A:

- Confidentiality
- Integrity
- Availability

❖ Expanded model adds:

- Accountability
- Nonrepudiation

❖ These are critical in wireless & mobile systems.

Confidentiality

❖ **Confidentiality means:**

- Preventing unauthorized disclosure
- Protecting data in storage and transit
- Encrypting mobile device data

❖ **Modern concern:**

- Data privacy
- Data secrecy
- Lost/stolen smartphones
- IoT usage tracking

❖ **Mobile devices increase confidentiality risks**

Integrity

❖ Integrity ensures:

- Data is genuine
- No unauthorized modification
- No tampering

❖ Tools used:

- Hash functions
- Message digests
- Even one-bit change → different hash value.

❖ Prevents:

- Man-in-the-middle attacks
- Session hijacking

Availability

❖ **Availability means:**

- Systems are accessible when needed
- Services are not denied

❖ **Security-related threats:**

- DoS attacks
- SYN floods
- Wireless jamming
- Backhaul disruption

❖ **Wireless adds new availability risks.**

SYN Flood Explained

❖ TCP handshake:

- SYN
- SYN-ACK
- ACK

❖ In SYN flood:

- Attacker sends many SYN requests
- Never completes handshake
- Server resources exhausted

❖ ***Result:*** Legitimate users blocked.

Accountability

❖ **Accountability ensures:**

- Users are responsible for actions
- Activities are logged
- Audit trails exist

❖ **Requires:**

- Authentication
- Logging
- Monitoring
- Authentication ≠ Authorization

Authorization vs Authentication

❖ Authentication:

- Verifies identity (username/password)

❖ Authorization:

- Determines access rights

❖ Example:

- Verified employee
- But restricted access based on role

❖ Lost device risk:

- Authorized device in wrong hands = disaster

Nonrepudiation

Nonrepudiation prevents denial of action.

❖ **Achieved using:**

- Digital signatures
- Cryptographic proof

❖ **Used in:**

- E-commerce
- Financial transactions
- Legal communications

❖ Wireless session hijacking can weaken nonrepudiation.

Threat Categories

Wireless & Mobile Threat Categories

❖ Three broad categories:

1. **Data theft**
2. **Device control**
3. **System access**

❖ All wired threats also exist in wireless.

❖ Wireless introduces additional dimensions.

Data Theft Threats

Data Theft Goals

❖ Hackers target:

- Credentials
- Personally Identifiable Information (PII)
- Remote access tools
- Corporate data
- LTE broadband devices

❖ Most attacks are opportunistic.

Sniffing / Snooping

❖ Radio signals:

- Easily intercepted
- No physical access required

❖ Solution:

- Encryption

❖ Even weak encryption:

- Stops casual attackers
- Raises attack cost

Malicious Applications (Malware)

❖ **Mobile app ecosystem:**

- Thousands of apps
- Increased attack surface

❖ **Risks:**

- Spyware
- GPS tracking
- Email copying
- Blackmail scenarios

❖ **Strict app control policies required.**

Browser Exploits & BYOD

❖ **BYOD introduces:**

- Unpatched browsers
- Delayed updates
- Vulnerable endpoints

❖ **Simply visiting malicious site:**

- Triggers exploit
- Installs malware

❖ **Mobile Device Management (MDM) is critical.**

Wireless Phishing & Evil Twins

❖ Phishing:

- Fake SMS/email links
- Fraudulent login pages

❖ Small screens:

- Harder to detect fraud

❖ Evil Twin:

- Fake access point
- Same SSID (Service Set Identifier; a unique name for a wifi network)
- Users connect unknowingly

Lost or Stolen Devices

Most common mobile threat.

❖ **Consequences:**

- Data exposure
- VPN misuse
- System access compromise

❖ **Best mitigation:**

- Strong passcodes
- Immediate reporting
- Remote wipe

IoT System Takeover

❖ **Wireless IoT risks:**

- Remote appliances shutdown
- Physical access control compromise

❖ IoT merges cyber and physical threats.

Device Control Threats

Lateral Movement (Lily Padding)

❖ Attackers:

- Compromise one device
- Move laterally
- Reach high-value targets

❖ Also called:

- Island hopping

❖ Device control is a stepping stone.

Jailbreaking & Rooting

❖ Users may:

- Modify devices
- Remove security restrictions
- Install unsigned applications

❖ Risks:

- Backdoors
- Corporate network compromise

❖ Major concern in BYOD.

Ad Hoc & Endpoint Attacks

❖ Threats include:

- Ad hoc networks
- Unauthorized hotspots
- Tools like Metasploit

❖ These tools:

- Probe for vulnerabilities
- Automate exploitation

Bluetooth & NFC Attacks

❖ Bluetooth:

- If in discovery mode → vulnerable

❖ NFC:

- Automatic pairing
- Used in payments
- Future attack vector

❖ Proximity does not equal security.

System Access Threats

DoS & Jamming

❖ **Wireless-specific DoS:**

- Radio interference
- Channel flooding
- Backhaul disruption

❖ 5 GHz reduces accidental interference.

❖ Targeted jamming still possible.

Rogue & Evil Twin Access Points

❖ Evil Twin:

- Fake SSID
- User connects unknowingly

❖ Rogue AP:

- Unauthorized employee setup
- Hacker-installed access point
- Regular site surveys required.

Summary

- ❖ Wireless expands attack surface.
- ❖ Security must align with business needs.
- ❖ C-I-A-A-N model is foundational.
- ❖ BYOD introduces new risks.
- ❖ Encryption and access control are essential.
- ❖ IoT adds physical-world consequences.
- ❖ Proactive monitoring is critical.

*Thank
you*

